

CSA1717-ARTIFICIAL INTELLIGENCE

PRACTICAL LAB EXPERIMENTS

PROLOG PROGRAM TO SUM INTEGERS FROM 1 TO N

Aim:

To write a Prolog program to calculate the sum of integers from 1 to N.

Objective:

To implement recursive addition in Prolog using facts, rules and arithmetic operations.

Algorithm:

1. Start.
2. Read the value of N.
3. If $N = 0$, return sum as 0.
4. Otherwise, recursively calculate the sum from 1 to $N-1$.
5. Add N to the recursive result.
6. Display the sum.
7. Stop.

Flowchart:

START

```

|
v
Read N
|
v
Is N = 0?
/      \
Yes      No
|        |
Sum=0 Sum(N-1)
|        |
|      Add N
|      | \
|      /
v      v
Display Sum
|
v
STOP

```

Prolog Program:

% Sum of integers from 1 to N

sum_to_n(0, 0).

sum_to_n(N, Sum) :-

 N > 0,

 N1 is N - 1,

 sum_to_n(N1, S1),

 Sum is N + S1.

Output:

```
54 ?- sum_to_n(10, S).  
S = 55
```

Result:

Thus, the Prolog program successfully calculates the sum of integers from 1 to N.